Housing First or Treatment First?

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Response to Referee 2

Conceptual and detail concerns, responses in red

Thank you for the report. Of the two welfare margins in your second paragraph, the second, that parents care about what becomes of their children, is now the centre of the paper. I answer the first at C2.

One note before the comments. The paper has a new title: it was submitted as *Housing First or Treatment First?* and is now *Housing First or Treatment First? Ask the Kids, Not the Parents*. The reason is the substance of the revision rather than a relabelling, and it will be clearest at C3, since the mechanism the title names is the one you asked for there.

What changed, in brief. The paper now asks whether housing assistance should require sobriety, within the rationed voucher programme that exists, and answers it with a calibrated general equilibrium life cycle model in which a parent's drug use harms its child and the parent values its child only in part. Moral hazard alone does not justify the condition. Without the externalities on children, a newborn is worse off under Treatment First, by 1.09 percent of lifetime consumption, even when the savings from the drug test buy vouchers for other households. With them, in the long run and with the drug price and the voucher budget both adjusting, Treatment First cuts drug use by 21 percent and homelessness by 11 percent, and a newborn gains 2.57 percent of lifetime consumption, three quarters of it an externality its parent does not count. The parents gain too, by 0.69 percent of their own (Sections 6.2 and 6.3, Tables 5 and 6). Along the transition, neither homelessness nor the share of children in a shelter rises (Section 6.4).

Several changes bear directly on your report. The parent's concern for its child, which you raised in C3, is the paper's central mechanism. The effect of housing on recovery from a substance use disorder (SUD) is calibrated to Milby et al. (2005), the response of drug use to the condition itself is set to match the US military's random drug testing, and the model comes close to the same trial's comparison of contingent with unconditional housing without being targeted (C1). Preferences are the same for every household, whatever its disorder, its mental health or its dwelling, and a new two period stylized model (Section 3) shows that it is the uninsurable risk of dying, not the utility function, that puts drug use at the bottom of the income distribution (C4, C5). Homeownership, house prices, endogenous rents and labour supply are gone, and the housing grid is a shelter and three rental sizes (D1). Four things you asked for are not in the paper, or are only partly in it, and I say so where they come up: the disamenity of homelessness is a robustness check rather than part of the benchmark (C2), the paper does not plot the drug use policy function by income and wealth (C4), it does not solve for the optimal voucher programme (C6), and it does not test a uniformly spaced rental ladder (D1).

Conceptual concerns

C1 The welfare results rest on the elasticity of drug use to housing access

> *"... all of the welfare results rest on whether the model has an accurate elasticity of drug use to housing access. The only empirical drug use elasticity mentioned is a price elasticity of drug use*

participation, but it isn't clear to me that this provides enough or the correct discipline for these policy experiments. I would like to see that the model matches some evidence on the response of drug use to housing access (see Milby, et al (2005) for cocaine users, but there are many papers on this topic.)"

You are right that a price elasticity is not the right discipline for a housing policy, and the revised paper no longer leans on one. The calibration does not target the price elasticities at all; they are now an untargeted check. The Associate Editor singled out the same elasticity you did. The response of drug use to housing is now disciplined in three places, with Milby et al. (2005) at the centre, as you suggested.

Housing and recovery. Housing reaches a household's drug use through recovery and through the value of life. On recovery, a household with a SUD is less likely to recover in a shelter than in a home, by the multiplier μ_H on the recovery hazard (Section 4.6, equation (18)). μ_H is calibrated to Milby et al.: among homeless adults with a cocaine disorder in treatment, those who were housed had 1.47 times the abstinence of those who were not, and the model gives 1.45 (Section 5.1, Block 4, and Table 3). Housing changes the chance of recovering, not what the drug is worth to anyone, which also bears on your C5. On the value of life, housing makes life worth more, and a household with more to lose from dying is deterred more by the mortality risk of a SUD (Section 5.1, Block 2).

The response to the condition itself. What the experiments turn on is how much drug use responds when the voucher is made conditional on not using, which is a response to housing access rather than to the price. The detection rate of Treatment First's screen is set so that the screen deters use as much as the closest evidence on a drug test with a severe penalty. After the US military began testing its personnel at random and dismissing those who failed, service members who stayed were 61 percent less likely to have used marijuana in the past year (Mehay and Pacula, 1999). In the model, with a detection rate of 20 percent, 62 percent of the users who face the screen stop when it is introduced (Section 6, "The screen"). The 61 percent is the raw comparison among those who stayed, and the decline is 11 to 45 percent once those who left are counted, so I also run the screen at half the rate. Then 41 percent of users stop, inside that range, Treatment First cuts drug use by 14 percent and homelessness by 6 percent in the long run, and a newborn still gains 1.65 percent of lifetime consumption, against 2.57 at the calibrated rate (footnote 35).

Two checks against trials, neither of them targeted. Milby et al. also randomised housing that was contingent on abstinence against housing that was not. The contingent arm reached 8.32 consecutive weeks of abstinence against 6.60, a ratio of 1.26. The model's counterpart, the recovery hazard of voucher holders with a SUD under Treatment First's condition against that of housed households with a SUD and no voucher, is 1.40 in the first year of the condition, the horizon of the trial (Section 5.2, Block 8). The same trial therefore calibrates the effect of housing and checks the effect of the condition. Costantini (2025) adds a check on what housing does to mortality rather than to drug use: enrolling in the Housing First programme of the Department of Veterans Affairs lowers the three year mortality of homeless veterans with mental illness by 4.6 percentage points. In the model, losing a voucher to Treatment First's revocation raises the three year mortality of households with serious mental illness (SMI) by 3.6 percentage points, the same order of magnitude (Section 5.2, Block 7). The paper makes this comparison for his Housing First estimate only; it does not compare the model with his finding on the Department's Treatment First programmes.

The wider literature. Most randomised trials of Housing First compare unconditional housing

with usual care. They find large gains in housing stability and mixed to null effects on drug use, and they follow participants for at most three years. Only one trial, Milby et al.'s, compares a sobriety condition with unconditional housing, which is the comparison the paper is about, so it is the one the model is held to (Sections 1 and 2).

The price elasticities. The model's participation elasticity is -1.32, larger in absolute value than the -0.3 to -0.9 in Saffer and Chaloupka (1999) and Grossman and Chaloupka (1998), and its intensive elasticity is -0.63, inside the -0.3 to -0.8 in Grossman and Chaloupka (1998) and Olmstead et al. (2015). A larger price response works against Treatment First, since the screen lowers the drug price and a cheaper drug draws users back, so in this respect the welfare gain is conservative (Section 5.2, Block 5).

In the paper: Section 4.6; Section 5.1, Block 4, and Table 3; Section 6, "The screen", and footnote 35; Section 5.2, Blocks 5, 7 and 8, and Table 4; Section 2.

C2 A large homeless population is a negative externality

> *"... a large homeless population, especially if many are addicted to drugs, creates a negative externality beyond the fiscal burden of providing services. ... One structural rental housing paper that considers the negative externality of homelessness (though doesn't have drug addiction) is Corbae, et al (2024). ... the externality seems very easy to introduce into the current model (though would require a new calibration moment)."*

[Still open.] I solved the model with this disamenity, but the welfare numbers in the paper leave it out, and I want to say so plainly since you asked for it. I priced it from a direct valuation rather than from a new calibration moment. In a contingent valuation, Loubière et al. (2020) find that Europeans would pay about 0.2 percent of average earnings a year to end homelessness. With a disamenity of that size, no long run gain in Table 5 moves by more than 0.14 percentage points. Since it barely changes the results, the paper reports it in a footnote and does not carry it in the benchmark.

On your point that the externality is larger when many of the homeless have a disorder: under Treatment First the homeless population is both smaller and less often has a SUD. In the main experiment, homelessness falls from 0.62 to 0.55 percent of households, and the share of the homeless with a SUD falls from 20.2 to 15.6 percent (Table 5).

The paper does not cite Corbae, Glover and Nattinger (2024). With the disamenity left to a footnote, I took its size from Loubière et al.'s valuation rather than from a structural model.

The loss of consortium adjustment in the previous version, which Referee 1 and the Associate Editor objected to, is gone, and nothing is added to welfare from outside the model. The margin that decides the comparison is the one you raised in C3: the externalities on children, which sit inside the parent's own problem through the weight it puts on its child.

In the paper: Section 2, footnote 4; Table 5.

C3 Parents suffer anxiety about their children

> *"... parents suffer anxiety that their children may become drug addicts and/or homeless in adulthood. ... the dynastic element seems like more of a stretch, but could be discussed (I think it would only amplify the welfare gains from reducing addiction)."*

[New result.] This is now what the paper is about, and it is why the title changed. You and the

Associate Editor, who objected that the previous version discussed intergenerational transmission without any link between generations in the model, pointed to the same missing piece, and the revised model puts it in.

How the parent values its child. Every household has a child, who lives with it for its first 25 years, while the parent is 26 to 50, shares its consumption and its dwelling but not the drug, and then forms a household of its own, with a child of its own (Sections 4.1 and 4.2). The parent values its child at the weight ω_c in two places (equation (24)): the child's flow utility while it lives at home, and the child's adult life, through the loss the parent's own disorder inflicts on it. The second term is a difference: the child's value at entry to adult life had the parent never had a SUD, less its value given the parent's actual history (equation (25)). It is zero for a parent who never had a SUD and positive otherwise. Written as a level, the term would be worth more to the households likelier to die, and so most to those with a SUD, for reasons that have nothing to do with the child (footnote 23).

This needs no dynasty in the state. A parent's choices reach its child's adult life through the child's productivity, which is drawn around the parent's and is lowered by 24 percent for life if the parent had a SUD at any point while the child lived at home (Section 4.8, equation (22)), the earnings loss that Nilsson (2017) measures. The lower draw makes the child likelier to be poor, to use, to have a SUD and SMI, to be homeless and to die earlier. So the parent's concern covers the two outcomes you name, a child who becomes addicted or homeless as an adult, to the extent that the parent's own choices can change them.

How much the parent cares. Altonji et al. (1997) reject perfect altruism, but their test needs transfers between a parent and its grown child, which the model does not have. So ω_c is set by behaviour. In the 2024 National Survey on Drug Use and Health (NSDUH), an unmarried adult living with one other adult and no child was 17.5 percent more likely to have used an illicit drug in the past year than a lone parent living with one child, with the lone parents reweighted to the other group's age, sex, marital status and family income. Every household in the model has a child, so the model's counterpart is a counterfactual: ω_c is the weight at which giving the child no weight raises drug use by as much, 17.7 percent in the model, and it is 0.085 (Section 5.1, Block 5, and Table 3). With the weight Nishiyama (2000) estimates, 0.69, at the upper end of the literature, Treatment First still raises a newborn's welfare, by 3.18 percent of lifetime consumption against 2.57. Less of the gain is then an externality, 23 percent of it against 76 percent (footnote 28).

Three externalities. Because the parent counts only the share ω_c, three costs of its drug use fall on the child without being fully weighed: the consumption and housing the child goes without when the parent spends on the drug or is out of work with a SUD (the budget externality); the earnings penalty (the earnings externality); and the shelter, since a child lives wherever its parent lives (the childhood homelessness externality). The welfare criterion counts the child's childhood at its full weight (Section 6.1).

What it does to the answer. You expected the dynastic element to amplify the welfare gains from reducing addiction. In this model it decides their sign. Without the externalities, the child appears only as an adult: its parent gives it no weight, its productivity does not depend on its parent's disorder, and its childhood is not counted. A newborn then loses under Treatment First, by 1.17 percent of lifetime consumption with the drug price and the voucher probability held, 1.09 percent with the voucher probability balancing the budget, and 0.74 percent with both adjusting. With the externalities it gains 2.82, 6.45 and 2.57 percent in the same three cases (Section 6.2, Table 5). In the main experiment the 2.57 percent splits into 1.50 percent from childhood and

0.46 percent from better productivity at entry, both of which the parent does not count, 0.43 percent from the newborn's own adult life under the policy, and 0.19 percent that the parent does count (Section 6.3, Table 6). The earnings externality is the smaller part, 18 percent of the gain, and Treatment First still raises welfare with the penalty set to zero (footnotes 5 and 30).

Childhood homelessness. The externality that critics of Treatment First fear runs the other way, through the shelter. In the main experiment the share of children in a shelter falls from 1.03 to 0.92 percent, and along the transition it never rises (Sections 6.2 and 6.4).

Whether it is a stretch. Users do have children. In the National Survey of Family Growth 2017 to 2019, 75 percent of adults aged 40 to 49 who used hard drugs in the past year had a child, against 81 percent of those who used none, and among adults aged 26 to 49, 40 percent of those with a drug use disorder have a child under 18 at home, against 48 percent of comparable adults who used no illicit drug. Even if all of the gain that comes in childhood shrank in proportion to that gap, the gain would be only 11 percent smaller (Section 2). In the model, 5.82 percent of children live with a parent who has a SUD, against 4.6 percent in NSDUH 2024, and a child of a parent with a SUD has 1.46 times the odds of acquiring one, at the low end of the 1.44 to 2.51 in Mellentin et al. (2016) and Yoon et al. (2013) (Section 5.2, Block 6).

In the paper: Sections 4.1, 4.2, 4.8 and 4.10; Section 5.1, Block 5, and Table 3; Section 5.2, Block 6; Sections 6.1 to 6.4, and Tables 5 and 6.

C4 The rich model is a black box; show the policy functions

> *"... it can be a bit of a black box. It would be helpful to have a section where the mechanisms of the model are explored more thoroughly. For example, it would be useful to show and explain intuitively a plot of the drug use policy function for addicts and non-addicts as their income and wealth vary."*

[Still open.] The paper does not include the plot you describe, of the drug use policy function by income and wealth for households with and without a SUD. In the quantitative model the decision to use depends on age, assets, productivity, employment, the disorder, SMI and the voucher, so I chose to show the mechanism where it can be isolated, in a stylized model, and to show the quantitative model's drug use by income against the data. The paper explores the mechanisms in three ways.

A stylized model. Section 3 is a new two period model, calibrated (Tables 1 and 2), whose sole purpose is to show why drug use concentrates at the bottom of the income distribution. Use there depends on productivity alone, so each consequence of a SUD can be scaled up and down with the other switched off, and its effect on who uses read off directly (Figure 1, with b recalibrated at every point so that the share who use stays at 8.90 percent). With no consequences, the average user would earn 3.51 times the population average. The risk of dying alone takes that to 0.64 at its calibrated strength. The risk of losing one's job alone leaves it at 3.51, because saving insures it; with saving switched off, the job risk alone takes it to 0.75. Section 3.3 gives the intuition, which I set out at C5.

Drug use by income in the quantitative model. Figure 2 is the closest the paper comes to your plot: the share of households that use drugs at each level of income relative to the mean, model against NSDUH 2024, averaged over the other states. As in the data, use is highest at the bottom and lowest at the top: 18.2 percent of households below 0.22 of mean income use, against 13.7 percent in the data, and 4.2 percent of those above 1.62 times the mean, against 4.8 percent. In between, use in the model is flatter and lower than in the data (Section 5.2, Block 2).

The experiments, one step at a time. Table 5 introduces Treatment First with one equilibrium condition at a time, the screen alone, then the voucher budget, then the drug price, with and without the externalities on children. Section 6.3 and Table 6 split the newborn's gain into four parts, two of them externalities, and say which the parent counts.

In the paper: Section 3 and Figure 1; Section 5.2, Block 2, and Figure 2; Sections 6.2 and 6.3, and Tables 5 and 6.

C5 The utility specification bakes in an income contingent gain or loss

> *"It would also help to further explain the importance of the period utility specification in (13) which, I think, implies that drug addiction is more attractive for low income people . . . it also bakes in an income-contingent gain/loss from becoming addicted by shifting the utility weight towards drug consumption away from non-drug consumption and housing."*

[Conceded.] You and Referee 3, in comment M2(a), identified the same problem with the previous specification. The weight on the drug rose when a household acquired the disorder, so reducing the disorder mechanically raised the utility of consumption and housing, and part of the welfare comparison came from the utility function rather than from behaviour.

In the revised model, preferences are the same for every household. There is one weight on the drug, θ, one weight on housing, γ, and one shifter, b, whatever a household's SUD status, its mental health or its dwelling (Section 4.2, equations (11) and (12)), and SMI does not enter utility at all (Section 4.7). The Associate Editor suggested keeping the weights fixed and letting b vary with the disorder; the revision keeps b fixed as well. The only utility term tied to the disorder is the cost of a year in treatment, which a household with a SUD chooses whether to pay. What a SUD costs runs through the risks it brings and through the child: a higher chance of dying, a lower chance of finding work, a higher chance of SMI, less consumption and housing for the child when the parent spends on the drug or is out of work, and a productivity penalty on the child, who inherits a worse draw, not the disorder or a taste for the drug (Sections 4.6 to 4.8).

On your question, whether the utility function makes drug use more attractive at low income: in the revised model it does the opposite. Drugs are a luxury good bought at a price. Because $b > 0$, the first unit has a finite marginal value, so most households buy none, and a household with more income gives up less utility for each dollar it spends on the drug. With no consequences of use, use would therefore concentrate at the top: in the stylized model, the average user would earn 3.51 times the population average (Section 3.3).

What reverses the gradient is the risk of dying, which cannot be insured. Dying takes away the whole continuation value of being alive, which grows with income, so it deters high income households and not low income ones. Losing a job is partly insured by saving, so its cost is roughly proportional to income and, on its own, does not change who uses. In the stylized model the risk of dying alone takes the average user from 3.51 times the population average to 0.64, and with both risks at their calibrated strength it is 0.74, as in NSDUH 2024 (Figure 1). So the cost of a SUD does depend on income, as you suspected, but through the mortality risk of the disorder rather than through preferences, and it is larger for richer households. In the quantitative model it is measured rather than assumed: the mortality multiplier of a SUD is calibrated to the income of users, 74 percent of the average in the data and 62 percent in the model, and its value, 4.00, is within the 3.91 to 6.6 range of mortality rate ratios in the literature (Section 5.1, Block 1). Figure 2 then compares the whole profile of use by income with the data.

Keeping preferences fixed is also what lets the paper explain why users are poor without assuming

that preferences differ systematically across households. Greenwood et al. (2025) reproduce the same fact by letting preferences for the drug differ by education; here no preference parameter varies with income (Section 2).

In the paper: Sections 4.2 and 4.7; Section 3.3 and Figure 1; Section 5.1, Block 1; Section 5.2, Block 2, and Figure 2.

C6 Optimal voucher design, and χ_H looks high

> *"The parameter χ_H governs the fiscal cost of having somebody unhoused and appears pretty high at 25% of average income. Would there be significant further welfare gains from broadening the voucher program by making higher incomes eligible or covering more of the rent (with the drug-free contingency)? My (perhaps wrong) reading is that the cost of having somebody unhoused is significantly higher than the rent on the smallest housing unit."*

[Still open.] Two things have changed, and the second means the paper does not answer your question directly.

The fiscal cost of homelessness. The revised model has no fiscal cost of homelessness. The shelter is free, and the government's budget carries the income taxes, the estates of the dead, the endowment of entrants, unemployment insurance, social security, food assistance, the voucher, the drug test and treatment, and nothing for a household in a shelter (Section 4.11, equation (26)). So the parameter you asked about is gone, and with it the comparison between the cost of a homeless household and the rent on the smallest unit. Leaving it out does not help Treatment First: Treatment First lowers homelessness in the long run and does not raise it along the main transition, so a cost per homeless household would only add to what it saves. What pays for more vouchers under Treatment First is instead the vouchers revoked from users who are caught, and the taxes of adults who start adult life without the earnings penalty or who are not out of work with a SUD. In the main experiment the probability of drawing a voucher rises from 6.75 to 7.80 percent (Section 6.2).

Broader programmes. The previous version compared programmes with and without rationing. The revision does not, and it does not solve for the optimal programme. It keeps the programme rationed and its rules at those of the Department of Housing and Urban Development (HUD), eligibility below half of median income and a tenant contribution of 30 percent of income. It holds the tax system and net government revenue at their benchmark levels, and asks only whether the assistance that exists should require sobriety (Section 6). Raising the eligibility cutoff or the share of the rent covered is not among the experiments, so the paper cannot tell you whether either would add further gains.

What the paper does show is the value of more vouchers that carry the sobriety condition, at the same net revenue. When the probability of drawing a voucher is free to spend what the screen saves, with the drug price held, it rises from 6.75 to 9.24 percent, homelessness falls by 20 percent and the share of children in a shelter by 19 percent, and a newborn gains 6.45 percent of lifetime consumption, against 2.82 percent when the savings go unspent. The parents gain 2.42 percent of their own, against a loss of 1.19 percent (Section 6.2, Table 5, seventh and eighth columns; Table 6).

In the paper: Section 4.11; Section 6; Section 6.2, and Tables 5 and 6.

Detail concerns

D1 The housing grid

> *"... the housing grid. It has seven values but only three are calibrated ... The full grid is then listed in footnote 4 rather than a calibration table. My assumption before seeing the grid was that the bounds were calibrated and then the sizes were distributed uniformly ... But they are not uniformly spread out, at least in levels or ratios. So, how are the housing levels being chosen and does it matter for results? Could they just be spread uniformly without affecting the calibration and policy conclusions? If not then I think they need to be included as parameters in the calibration."*

[Conceded.] **[Still open.]** You were right that the previous grid had no principle behind its spacing. The revision makes the grid smaller and puts every point of it in a calibration table, but it does not spread the upper rungs uniformly or test whether their spacing matters, and I want to be clear about that.

The grid is now four points. Homeownership is gone, along with house prices, endogenous rents and labour supply, which in earlier versions had a negligible effect on the results (footnote 19). A household chooses between a shelter, h_1, and three rental sizes, $h_2 < h_3 < h_4$, at a fixed rent per unit (Sections 4.4 and 4.5).

The two points at the bottom are calibrated. They decide what a voucher, and its removal, are worth to a poor household. The shelter's service flow h_1 is set so that homelessness is within 0.40 to 1.00 percent of households, the floor from HUD's 2023 point in time count of a single night and the ceiling from the share homeless at some point over a year in Burt et al. (2001). The model gives 0.62 percent, with h_1 at 53 percent of the smallest rental (Section 5.1, Block 2, and Table 3). The smallest rental h_2 decides how much of its income a poor household must spend to stay out of the shelter. It is set to the share of renters paying more than half of their income for housing, 19.2 percent in the 2022 Survey of Consumer Finances against 25.8 percent in the model (Appendix A, Block 8).

The two upper rungs are set. They are at 4.00 and 8.00 times the smallest rental, and they now appear as parameters in Table A1 rather than in a footnote. They are not calibrated to a moment, and they are not uniform in levels or in ratios. They decide what a household that can afford more than the smallest unit rents, and the paper does not report results with a different spacing. So the paper does not answer your last question, whether a uniform ladder would change the calibration or the conclusions.

In the paper: Sections 4.4 and 4.5; Section 5.1, Block 2, and Table 3; Appendix A, Block 8, and Table A1.